

**5 (a)** Write down the fourth roots of unity. [1]

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(b) Use de Moivre's theorem to show that

$$\cos 4\theta = 8 \cos^4 \theta - 8 \cos^2 \theta + 1. \quad [4]$$

[illegible]

(c) Hence obtain the real roots of the equation

$$16(8x^4 - 8x^2 + 1)^4 - 9 = 0$$

in the form  $\cos(q\pi)$ , where  $q$  is a rational number.

[5]

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